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RESEARCH ARTICLE

Associations Between Brain Metabolites Measured With MR Spectroscopy and Head Impacts in High School American Football Athletes

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Background: While changes in brain metabolites after injury have been reported, relationships between metabolite changes and head impacts are less characterized.
Purpose: To investigate alterations in neurochemistry in high school athletes as a function of head impacts, concussion, and the use of a jugular vein compression (JVC) collar.
Study Type: Prospective controlled trial.
Subjects: A total of 284 male American football players, divided into JVC collar and noncollar groups; 215 included in final analysis (age = 15.9 ± 1.0 years; 114 in collar group).
Field Strength/Sequence: 3 Tesla/T_1-weighted gradient echo, ^{1}H point resolved spectroscopy, acquired between August and November 2018.
Assessment: Head impacts were quantified using accelerometers. Concussion was diagnosed by medical professionals for each team. Pre- to postseason differences in total *N*-acetylaspartate (tNAA), total choline (tCho), myo-inositol (myoI), and glutamate + glutamine (Glx), in primary motor cortex (M1) and anterior cingulate cortex (ACC), relative to total creatine (tCr), were determined.
Statistical Tests: Group-wise comparisons were performed using Wilcoxon signed-rank, Friedman's, and Mann–Whitney *U* tests. Relationships between Δmetabolite/tCr and mean *g*-force were analyzed using linear regressions accounting for concussion and JVC collar. Significance was set at $P \leq 0.05$.
Results: In participants without concussion, a significant decrease in tCho/tCr ($0.233 \pm 1.40 \times 10^{-3}$ to $0.227 \pm 1.47 \times 10^{-7}$) and increase in Glx/tCr ($1.60 \pm 8.75 \times 10^{-3}$ to $1.63 \pm 1.08 \times 10^{-2}$) in ACC were observed pre- to postseason. The relationship between ΔtCho/tCr in M1 and ACC and mean *g*-force from >80 *g* to >140 *g* differed significantly between participants with and without concussion (M1 β ranged from 3.9×10^{-3} to 2.1×10^{-3}; ACC β ranged from 2.7×10^{-3} to 2.1×10^{-3}). Posthoc analyses revealed increased tCho/tCr in M1 was positively associated with mean *g*-force >100 *g* ($\beta = 3.6 \times 10^{-3}$) and >110 *g* ($\beta = 2.9 \times 10^{-3}$) in participants with concussion. Significant associations between ΔmyoI/tCr in ACC and mean *g*-force >110 *g* ($\beta = -1.1 \times 10^{-3}$) and >120 *g* ($\beta = -1.1 \times 10^{-3}$) were observed in the collar group only.

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Data Conclusion: Diagnosed concussion and the use of a JVC collar result in distinct neurochemical trends after repeated head impacts.
Level of Evidence: 2
Technical Efficacy: Stage 3

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Mild traumatic brain injury (TBI) is a frequent occurrence in contact sports.[1] The Centers for Disease Control and Prevention estimate ∼135,000 athletes with concussion, a form of mild TBI, are admitted to emergency departments annually in the United States.[2] Over the past two decades, increasing numbers of concussions have been observed in high school and collegiate athletes due to, in part, improved reporting and increased participation in sports.[1,3] Among contact sports, American football typically reports one of the highest rates of concussion.[2,4] During practices and games, athletes experience head impacts of varying number and magnitude which can lead to neurological impairment.[5] As the diagnosis of concussion is typically based on symptoms, the effect of head impacts across a range of forces is unclear,[5–7] and the heterogenous nature of both injury and clinical presentation remains a challenge.[8]

Noninvasive neuroimaging methods using MRI provide an objective way to quantify neurological alterations following concussion, which have been widely demonstrated in athletes.[9,10] While many studies have focused on brain structure[11] and connectivity,[12] changes in neurochemistry also occur after brain injury.[13–15] MR spectroscopy (MRS) enables noninvasive quantification of brain metabolite concentrations and can be easily incorporated into MRI protocols.[13,16] In the context of concussion, brain metabolites of interest include *N*-acetylaspartate (NAA), associated with neuronal integrity; choline (Cho), related to membrane turnover and cell proliferation; myo-inositol (myoI), an osmolyte linked to glial cell growth and neuroinflammation; and glutamate + glutamine (Glx), brain metabolites associated with immune modulation and neurotransmission, respectively.[10,13,17,18] Previous work using ^{1}H MRS reported lower total NAA + *N*-acetyl-aspartyl-glutamate (NAAG) (tNAA) relative to total creatine + phosphocreatine (tCr) in individuals with a diagnosed concussion ∼10 hours postinjury compared to healthy controls.[19] Additional longitudinal investigations of athletes participating in contact sports without a diagnosed concussion reported lower myoI at mid- or postseason compared to preseason.[20,21] These results suggest alterations in brain metabolites may be a quantitative metric for brain injury even in the absence of diagnosed concussion; however, reported trends in neurochemical changes after concussions are inconsistent.[10]

While repeated concussions are known to be detrimental to brain health, there is growing concern about the effect of repeated head impacts with high acceleration, even in the absence of a concussion.[7,22] To mitigate the effects of repeated head impacts, prior research indicates wearing a jugular vein compression (JVC) collar leads to venous dilation and reduces the space between brain tissue and skull, potentially mitigating damage during and after head impact.[23–25] Prior investigations reported longitudinal white matter alterations in American football athletes who did not wear a JVC collar, whereas no changes were observed in athletes wearing a collar, suggesting it may be neuroprotective.[11,24] Histological analysis in a rodent model of induced brain injury revealed the use of a JVC collar lowered astrocyte reactivity and microglial activation compared to a noncollar control group.[25] As cellular level damage can result in measurable changes in neurochemistry,[16] quantification of brain metabolites may provide insight into the potential benefits of the use of a JVC collar after head impact.

The aims of this study were to quantify changes in brain metabolites during a single season of American football with and without the presence of a JVC collar, as well as between individuals with and without a diagnosed concussion. We hypothesize brain metabolite concentrations vary between pre- and postseason, and repeated head impacts, concussion, and use of a JVC collar are associated with metabolite changes.

Materials and Methods

Subject Characteristics and Study Design

This prospective cohort study was approved by the local Institutional Review Board and registered with clinicaltrials.gov (NCT04068883). The Strengthening the Reporting of Observational studies in Epidemiology (STROBE) checklist is included in Appendix S1 in the Supplemental Material. Written informed consent was obtained from all participants, or their parent or guardian if the participant was under 18 years old. All participants under 18 years old provided written assent prior to participation. Male high school football athletes between 13 and 18 years old were recruited from seven greater Cincinnati area high schools. Participants were assigned to either a JVC collar group (four teams) or noncollar group (three teams). Participants in the collar group wore the JVC collar (Q-collar, Q30 Innovations, Westport, CT, USA) during practices and games throughout the competitive season (from early August to late November 2018, depending on the regular season schedule). MR data were acquired at two or three timepoints: before the regular season (preseason), after the last game (postseason), and (if applicable) approximately 1 week after a concussive event (postconcussion). The high school athletic trainer evaluated any athlete for whom a concussive event had been suspected by following

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standard clinical assessment guidelines (using the sport concussion assessment tool [SCAT]).[26] If there was a positive assessment for a concussion, the athlete was removed from play and referred to the school medical provider for diagnosis confirmation. Concussion diagnosis and management was handled clinically by the medical providers (athletic trainer/team physician) for the school based upon their standard of care. Postconcussion MR data were reviewed by a neuroradiologist (>24 years of experience). Exclusion criteria included a lack of medical clearance to play competitive sports, history of neurological deficits, cerebral infarction or severe head trauma, and contraindications to MRI.

Head Impact Tracking

Head impacts during the season were quantified using an accelerometer (CSx System Ltd., Auckland, New Zealand) affixed below each participant's left mastoid.[24,27–29] Impacts exceeding 10 *g* triggered recording for the next 45 msec, and peak *g*-force during this interval was used as the magnitude of the impact. To reduce false positives in head impact data, impacts were removed when ≥3 consecutive impacts >20 *g* occurred <10 seconds apart within a 30-second time frame, as these were considered likely to be spurious and unrealistic to impacts sustained during football.[24,27–29] These largely occurred at the beginning or end of each recording and were attributed to removal of the sensor or helmet. This resulted in removal of 28.5% of the total impacts recorded across all participants. Thresholds from >20 *g* to >150 *g* in 10 *g* increments were applied to head impacts for each participant and were used to calculate mean *g*-force of impacts (the sum of total *g*-force divided by the number of impacts at each threshold). All head impacts greater than or equal to each threshold were included for calculating the mean *g*-force (i.e., impacts below the threshold were excluded). This approach for thresholding was used to characterize the effects of cumulative head impacts above a *g*-force threshold.

MRI and MRS Acquisition and Analysis

MR data were acquired on three 3 Tesla Philips MR scanners (Achieva, Ingenia, and Elition) using a 32-channel phased-array head coil and identical sequence parameters across scanners. Each participant completed all MR examinations on the same scanner. A sagittal T_1-weighted image, used to position the MRS voxel, was acquired with a three-dimensional, gradient recalled echo sequence (repetition time [TR] = 8.1 msec, echo time [TE] = 3.7 msec, inversion time [TI] = 1070 msec, field of view [FOV] = 256 × 256 mm^2, voxel size = 1 mm isotropic, number of slices = 180). Single voxel ^{1}H MRS data were acquired in the left primary motor cortex (M1) and rostral anterior cingulate cortex (ACC) using a point resolved spectroscopy sequence (TR = 2000 msec, TE = 30 msec, averages = 96, complex data points = 1024, voxel size = 2 cm isotropic, bandwidth = 2000 Hz, flip angle = 90°). Water-suppressed (using chemical shift selective saturation suppression pulses) and nonwater-suppressed spectra were acquired. MR spectra were analyzed in an automated and blinded manner using LCModel v6.3-*I* with a GAMMA-simulated metabolite basis set.[30] Metabolites of interest included tNAA, total phosphocholine + glycerophosphocholine (tCho), myoI, and Glx, relative to tCr, which served as an internal reference to minimize confounding effects including MR scanner differences and repeated acquisition across multiple days. Relative tissue fractions (gray matter, white matter, cerebral spinal fluid [CSF]) of each MRS voxel were determined from tissue probability maps calculated using the Statistical Parametric Mapping toolbox (SPM12; Wellcome Trust Center for Neuroimaging, London, UK, https://www.fil.ion.ucl.ac.uk/spm/). Metabolite concentrations were corrected for CSF fractions, and tCr concentrations normalized to water were also corrected for T_1 and T_2 relaxation times of metabolites in gray and white matter.[31] The Dice similarity coefficient (DSC) for preseason and postseason voxel positions was calculated as two times the overlapped area divided by the total area in both voxels for each subject.[32] Spectra were excluded for voxels with DSC <0.5. The metabolite threshold for acceptable Cramer-Rao lower bounds was <20%, determined a priori. A Minimum Reporting Standards for in vivo Magnetic Resonance Spectroscopy (MRSinMRS) checklist[33] is included in Appendix S1 in the Supplemental Material.

Statistical Analysis

All analyses were performed using MATLAB (v2020b, Mathworks). Values are reported as the median ± standard error (SE) throughout unless otherwise noted. False discovery rate (FDR)-corrected *P*-values are reported for group-wise comparisons, with statistical significance set a priori at $P \leq 0.05$ for all analysis. While metabolites were generally normally distributed from visual inspection, due to slightly skewed distributions for some metabolites and the use of composite metabolite concentrations (eg, Glx), nonparametric statistical tests were used for group-wise comparisons. The appropriateness of tCr as an internal reference was evaluated using nonparametric Wilcoxon signed-rank tests to compare tCr between pre- and postseason. For participants without a diagnosed concussion, nonparametric Wilcoxon signed-rank tests were used to evaluate the differences in metabolite/tCr between preseason and postseason. For participants with a diagnosed concussion, nonparametric Friedman's tests were used to evaluate differences in metabolite/tCr between preseason, postconcussion, and postseason. The normalized change in metabolite/tCr was used for remaining analyses and denoted as Δmetabolite/tCr, where Δ represents (postseason − preseason)/preseason. Differences in Δmetabolite/tCr for all four metabolites between participants with and without a diagnosed concussion and between JVC collar and noncollar groups were evaluated with nonparametric Mann–Whitney *U* tests. A nonparametric two-sample Kolmogorov–Smirnov test was used to assess whether the distribution of head impacts differed between collar and noncollar groups.

Univariate linear regressions were used to assess the relationship between Δmetabolite/tCr (response), mean *g*-force of impacts for all thresholds from 20 *g* to 150 *g* (predictor), concussion (0 or 1, covariate), and the use of a JVC collar (0 or 1, covariate). The effects of high school and MR scanner were evaluated as covariates in the initial regression model but were not included in the final model as they were not significant for any *g*-force threshold or metabolite. The interaction effects of mean *g*-force of impacts × concussion and mean *g*-force of impacts × collar were assessed to identify changes in relationships as a function of concussion and collar, respectively. If significant interaction effects were observed, post-hoc univariate regressions were performed. To evaluate differences in associations between Δmetabolite/tCr and head impacts acquired prior to and after diagnosed concussion, linear mixed-effects modeling was used with Δmetabolite/tCr (response), mean *g*-force of impacts at each threshold (fixed effect predictor), preconcussion or postconcussion (categorical covariate), and subject (random effect for

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slope and intercept). The Akaike information criterion was used for mixed model selection. Three-way interaction effects of mean *g*-force of impacts × concussion × collar were evaluated for relationships between Δmetabolite/tCr and mean *g*-force of impacts where the interaction of impacts × collar was significant to determine if the observed effect was due, in part, to concussion.

Results

A total of 284 male subjects were recruited (mean ± standard deviation [SD] age = 15.8 ± 1.1 years) (Fig. 1). Of these, 215 unique participants (mean ± SD age = 15.9 ± 1.0 years; 114 in the collar group; 101 in the noncollar group) completed both pre- and postseason MRS scans, acquired approximately 4 months apart (Table 1). Voxel positioning and corresponding MR spectra for both voxels are shown in Fig. 2. One MR spectrum acquired in the ACC was excluded after visual assessment, and no metabolites for either voxel were excluded due to CRLBs. The median ± SE of the DSC between pre- and postseason voxels for M1 was 0.916 ± 0.005 (interquartile range [IQR] = 0.880–0.940), and for the ACC was 0.917 ± 0.005 (IQR = 0.881–0.935). During the season, 24 participants were diagnosed with a single concussion whose MRS data were collected approximately 1 week (mean ± standard deviation [SD] = 7.3 ± 2.4 days) after the concussive event. No participants were diagnosed with more than one concussion, and no visible changes or lesions were observed on structural images for any participant. The median ± SE number of head impacts (>20 *g*) during the season was 296 ± 20. No clear trends in competition level were observed between collar versus noncollar groups; both included the full range (divisions I–IV).

Differences in Metabolites Between Pre- and Postseason, Between Participants With and Without a Diagnosed Concussion and Between the Collar and Noncollar Groups

No significant differences in pre- to postseason tCr relative to water were observed in M1 (preseason: 6.95 ± 0.03, postseason: 6.94 ± 0.03, P = 0.55) or ACC (preseason:

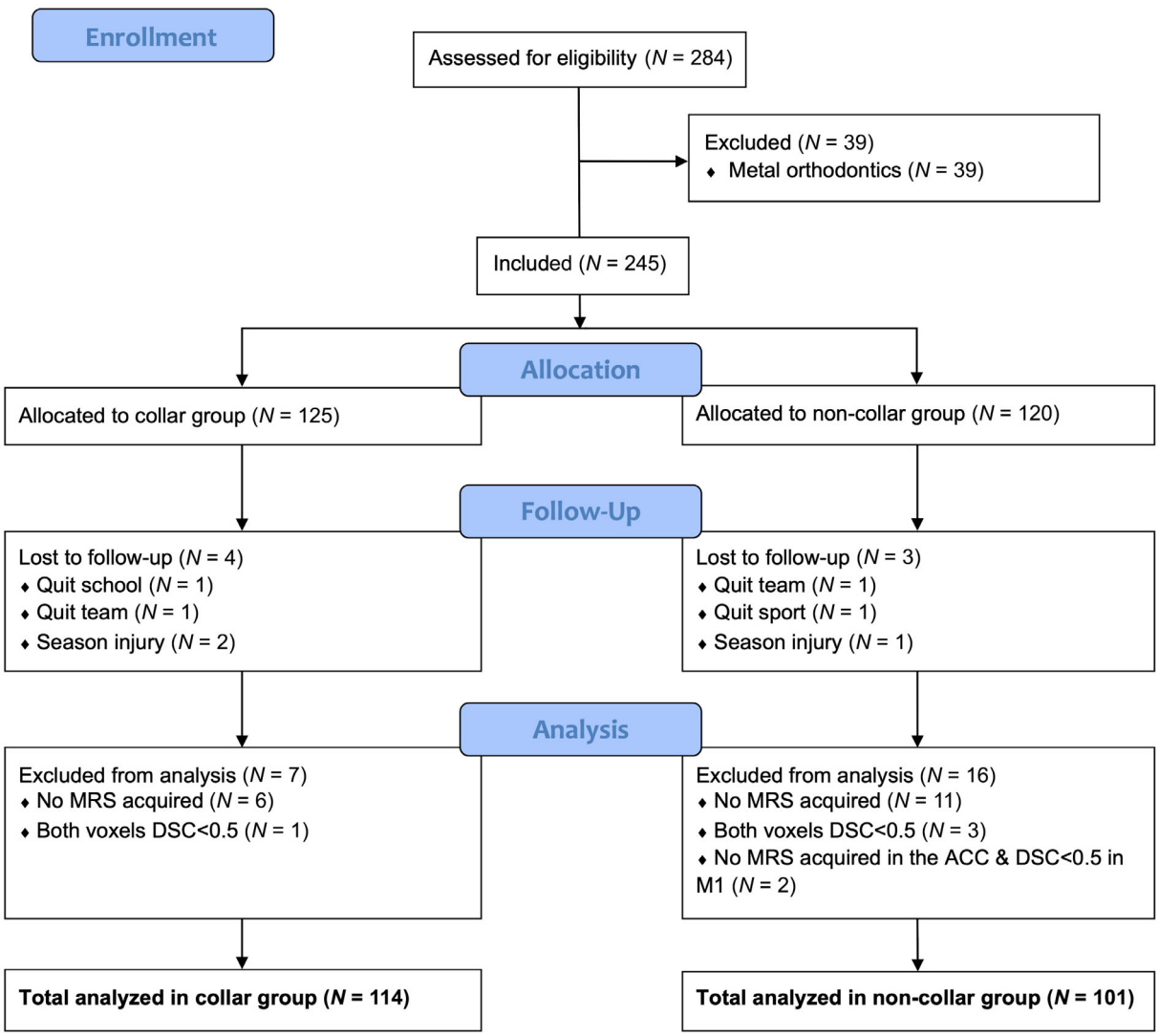

Figure 1: STROBE flow chart for enrollment, allocation, and inclusion and exclusion of clinical trial participants.

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Table 1. Description of Participants Included in the Final Analysis

Characteristics	All Participants (% of Total)	JVC Collar Group (% of Group)	Noncollar Group (% of Group)
Age (years)[a]	15.9 ± 1.0	16.0 ± 0.9	15.8 ± 1.1
Sex			
Male	215 (100)	114 (100)	101 (100)
Female	0 (0)	0 (0)	0 (0)
Diagnosed concussion	24 (11)	14 (12)	10 (10)

JVC = jugular vein compression.
[a]Data are mean ± standard deviation.

6.96 ± 0.03, 6.93 ± 0.04, $P = 0.36$), motivating its use as an internal reference for remaining metabolites. In participants without diagnosed concussion, a significant decrease of tCho/tCr (preseason: $2.33 \times 10^{-1} \pm 1.40 \times 10^{-3}$; postseason: $2.27 \times 10^{-1} \pm 1.47 \times 10^{-3}$) and an increase of Glx/tCr (preseason: $1.60 \pm 8.75 \times 10^{-3}$, postseason: $1.63 \pm 1.08 \times 10^{-2}$) were observed in the ACC (FDR-corrected) from pre- to postseason. No significant differences between preseason and postseason for tNAA/tCr (preseason: $1.50 \pm 7.50 \times 10^{-3}$; postseason: $1.50 \pm 7.45 \times 10^{-3}$; $P_{FDR} = 0.26$), tCho/tCr (preseason: $1.88 \times 10^{-1} \pm 1.33 \times 10^{-1}$; postseason: $1.88 \times 10^{-1} \pm 1.32 \times 10^{-3}$; $P_{FDR} = 0.12$), myoI/tCr (preseason: $6.66 \times 10^{-1} \pm 3.59 \times 10^{-3}$; postseason: $6.57 \times 10^{-1} \pm 3.87 \times 10^{-3}$; $P_{FDR} = 0.057$), or Glx/tCr (preseason: $1.08 \pm 7.45 \times 10^{-3}$; postseason: $1.07 \pm 7.83 \times 10^{-3}$; $P_{FDR} = 0.15$) in M1 were observed. Similarly, no significant differences between preseason and postseason tNAA/tCr (preseason: $1.11 \pm 8.17 \times 10^{-3}$; postseason: $1.12 \pm 7.85 \times 10^{-3}$; $P_{FDR} = 0.28$) or myoI/tCr (preseason: $7.38 \times 10^{-1} \pm 4.76 \times 10^{-1}$; postseason: $7.37 \times 10^{-1} \pm 4.34 \times 10^{-3}$; $P_{FDR} = 0.074$) in the ACC were observed (Table S1).

No significant differences between preseason, postconcussion, and postseason for tNAA/tCr (preseason: $1.52 \pm 1.55 \times 10^{-2}$; postconcussion: $1.53 \pm 1.64 \times 10^{-2}$; postseason: $1.52 \pm 1.08 \times 10^{-2}$; $P_{FDR} = 0.57$), tCho/tCr (preseason: $1.87 \times 10^{-1} \pm 3.34 \times 10^{-3}$; postconcussion: $1.81 \times 10^{-1} \pm 3.24 \times 10^{-3}$; postseason: $1.83 \times 10^{-1} \pm 4.04 \times 10^{-3}$; $P_{FDR} = 0.82$), myoI/tCr (preseason: $6.57 \times 10^{-1} \pm 1.08 \times 10^{-2}$; postconcussion: $6.63 \times 10^{-1} \pm 1.24 \times 10^{-2}$; postseason: $6.51 \times 10^{-1} \pm 1.51 \times 10^{-2}$; $P_{FDR} = 0.82$), or Glx/tCr (preseason: $1.05 \pm 2.67 \times 10^{-2}$; postconcussion: $1.11 \pm 1.63 \times 10^{-2}$; postseason: $1.10 \pm 1.60 \times 10^{-2}$; $P_{FDR} = 0.57$) in M1, and tNAA/tCr (preseason: $1.12 \pm 2.08 \times 10^{-2}$; postconcussion: $1.12 \pm 2.23 \times 10^{-2}$; postseason: $1.10 \pm 1.99 \times 10^{-2}$; $P_{FDR} = 0.36$), tCho/tCr (preseason: $2.25 \times 10^{-1} \pm 4.45 \times 10^{-3}$; postconcussion: $2.30 \times 10^{-1} \pm 3.73 \times 10^{-3}$; postseason: $2.24 \times 10^{-1} \pm 3.90 \times 10^{-3}$; P_{FDR} >0.99), myoI/tCr (preseason: $7.41 \times 10^{-1} \pm 1.13 \times 10^{-2}$; postconcussion: $7.55 \times 10^{-1} \pm 1.49 \times 10^{-2}$; postseason: $7.15 \times 10^{-1} \pm 1.49 \times 10^{-2}$; $P_{FDR} = 0.70$), or Glx/tCr (preseason: $1.64 \pm 2.73 \times 10^{-2}$; postconcussion: $1.65 \pm 2.50 \times 10^{-2}$; postseason: $1.63 \pm 3.10 \times 10^{-2}$; $P_{FDR} = 0.70$) in the ACC were observed in participants with a diagnosed concussion (Table S2).

Normalized changes in metabolite concentrations, Δmetabolite/tCr (Δ represents [postseason − preseason]/preseason) for all four metabolites in M1 and the ACC were not significantly different between participants with or without a diagnosed concussion (in M1, ΔtNAA/tCr: no concussion: $8.80 \times 10^{-3} \pm 4.11 \times 10^{-3}$, concussion: $8.41 \times 10^{-3} \pm 8.12 \times 10^{-3}$, $P_{FDR} = 0.98$; ΔtCho/tCr: no concussion: $8.06 \times 10^{-3} \pm 5.36 \times 10^{-3}$, concussion: $-1.81 \times 10^{-2} \pm 1.82 \times 10^{-2}$, $P_{FDR} = 0.78$; ΔmyoI/tCr: no concussion: $-6.85 \times 10^{-3} \pm 4.48 \times 10^{-3}$, concussion: $-1.87 \times 10^{-2} \pm 1.57 \times 10^{-2}$, $P_{FDR} = 0.78$; ΔGlx/tCr: no concussion: $-5.56 \times 10^{-3} \pm 6.29 \times 10^{-3}$, concussion: $2.05 \times 10^{-2} \pm 2.32 \times 10^{-2}$, $P_{FDR} = 0.78$; in the ACC, ΔtNAA/tCr: no concussion: $3.25 \times 10^{-3} \pm 6.53 \times 10^{-3}$, concussion: $-3.39 \times 10^{-2} \pm 1.96 \times 10^{-2}$, $P_{FDR} = 0.27$; ΔtCho/tCr: no concussion: $-1.39 \times 10^{-2} \pm 5.02 \times 10^{-3}$, concussion: $4.01 \times 10^{-3} \pm 1.58 \times 10^{-2}$, $P_{FDR} = 0.48$; ΔmyoI/tCr: no concussion: $-5.01 \times 10^{-3} \pm 6.06 \times 10^{-3}$, concussion: $-3.02 \times 10^{-2} \pm 1.48 \times 10^{-2}$, $P_{FDR} = 0.42$; ΔGlx/tCr: no concussion: $1.64 \times 10^{-2} \pm 6.43 \times 10^{-3}$, concussion: $-9.77 \times 10^{-3} \pm 1.76 \times 10^{-2}$, $P_{FDR} = 0.40$) (Table S3). Similarly, the use of a JVC collar had no significant effect on any Δmetabolite/tCr in M1 or in the ACC (In M1, ΔtNAA/tCr: collar: $9.88 \times 10^{-3} \pm 4.74 \times 10^{-3}$, noncollar: $8.80 \times 10^{-3} \pm 5.92 \times 10^{-3}$, $P_{FDR} = 0.62$; ΔtCho/tCr: collar: $1.06 \times 10^{-2} \pm 7.23 \times 10^{-3}$, noncollar: $1.92 \times 10^{-3} \pm 7.35 \times 10^{-3}$, $P_{FDR} = 0.62$; ΔmyoI/tCr: collar: $-5.72 \times 10^{-3} \pm 6.02 \times 10^{-3}$, noncollar: $-1.30 \times 10^{-2} \pm 6.29 \times 10^{-3}$, $P_{FDR} = 0.82$; ΔGlx/tCr: collar: $-8.10 \times 10^{-3} \pm$

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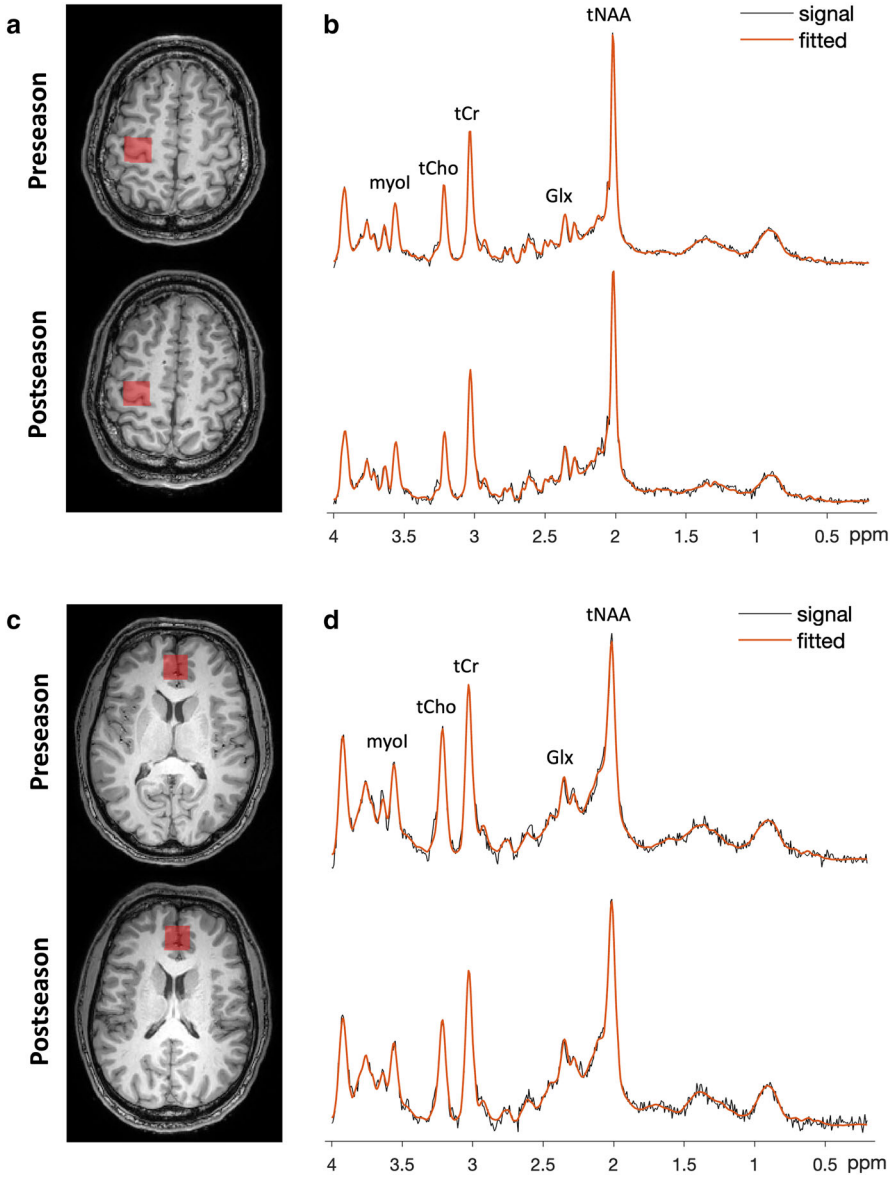

Figure 2: Representative magnetic resonance spectroscopy (MRS) voxel position for one subject in the left primary motor cortex (M1) and the rostral anterior cingulate cortex (ACC) with corresponding MR spectra. Spectra were fitted using LCModel. Images are shown in radiological orientation. (a) Pre- and postseason MRS voxel position in M1 (red box) overlaid onto T1-weighted images and (b) corresponding MR spectra (full width at half maximum [FWHM] = 0.031 ppm, signal-to-noise ratio [SNR] = 36). (c) Pre- and postseason MRS voxel position in the ACC overlaid onto T1-weighted images and (d) corresponding MR spectra (FWHM = 0.046 ppm, SNR = 20). (tNAA = total *N*-acetylaspartate + *N*-acetyl-aspartyl-glutamate; tCho = total phosphocholine + glycerophosphocholine; myoI = myo-inositol; Glx = glutamate + glutamine).

8.74×10^{-3}, noncollar: $-2.47 \times 10^{-3} \pm 8.69 \times 10^{-3}$, $P_{FDR} = 0.72$; in the ACC, ΔtNAA/tCr: collar: $1.31 \times 10^{-2} \pm 8.64 \times 10^{-3}$, noncollar: $-1.14 \times 10^{-2} \pm 8.95 \times 10^{-3}$, $P_{FDR} = 0.40$; ΔtCho/tCr: collar: $-1.29 \times 10^{-2} \pm 6.54 \times 10^{-3}$, noncollar: $-1.52 \times 10^{-2} \pm 7.10 \times 10^{-3}$, $P_{FDR} = 0.93$; ΔmyoI/tCr: collar: $-1.26 \times 10^{-2} \pm 8.14 \times 10^{-3}$, noncollar: $-5.06 \times 10^{-3} \pm 7.85 \times 10^{-3}$, $P_{FDR} = 0.93$; ΔGlx/tCr: collar: $8.51 \times 10^{-3} \pm 8.77 \times 10^{-3}$, noncollar: $1.04 \times 10^{-2} \pm 8.44 \times 10^{-3}$, $P_{FDR} = 0.93$) (Table S4).

Trends in Mean g-Force of Impacts and Concussion Rates

There were no clear trends in the median number of impacts >80 *g* (Fig. S1). The distribution of mean *g*-force of head impacts >80 *g* for participants in collar and noncollar groups was not significantly different (Kolmogorov–Smirnov test statistic [*D*] = 0.16, *P* = 0.10) (Fig. S2). No significant trends were observed between the number of head impacts >80 *g* for participants with and without a diagnosed concussion (concussion: 25.0 ± 3.0, no concussion: 24.5 ± 1.5, *P* = 0.54)

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(Fig. S3). The >80 *g* threshold was used as a median value for quantifying impact trends, but similar trends were qualitatively observed for the other *g*-force thresholds. For participants with diagnosed concussion, no clear trends in concussion timing were observed across the season (Fig. S4). The effects of high school and MR scanner were not significant for any metabolites or mean *g*-force of any thresholds and were excluded from the final regression model.

Relationships Between Metabolite Changes in the M1 and Mean g-Force of Impacts

A schematic for the calculation of mean *g*-force of head impacts is shown in Fig. 3. In M1, the relationship between ΔtCho/tCr and mean *g*-force of impacts varied significantly between participants with and without a diagnosed concussion for thresholds between >80 *g* and >140 *g* (>80 *g*: $\beta = 3.1 \times 10^{-3}$, >90 *g*: $\beta = 2.9 \times 10^{-3}$; >100 *g*: $\beta = 3.9 \times 10^{-3}$; >110 *g*: $\beta = 3.3 \times 10^{-3}$; >120 *g*: $\beta = 2.8 \times 10^{-3}$; >130 *g*: $\beta = 2.3 \times 10^{-3}$; >140 *g*: $\beta = 2.1 \times 10^{-3}$) but not for thresholds below 80 *g* or >150 *g* (Fig. 4a). Relationships between Δmetabolites/tCr in M1 and head impacts did not significantly vary for any metabolite between collar and noncollar groups. Posthoc regressions revealed significant positive associations between ΔtCho/tCr in M1 and head impacts for mean *g*-force thresholds of >100 *g* ($\beta = 3.6 \times 10^{-3}$) and >110 *g* ($\beta = 2.9 \times 10^{-3}$) in participants with a diagnosed concussion (Table 2). Within the concussion cohort, a significant interaction effect of mean *g*-force × time of injury was observed for thresholds of >100 *g* ($\beta = -4.1 \times 10^{-3}$) and >110 *g* ($\beta = -3.5 \times 10^{-3}$), suggesting the trend between changes in tCho/tCr in M1 and head impacts prior to the diagnosed concussion (between preseason and postconcussion) differed from the trend after the concussion (between postconcussion and postseason) (Fig. 5). Relationships between Δmetabolite/tCr in tNAA, myoI, and Glx in M1 and head impacts with any mean *g*-force thresholds were not significant and did not vary between participants with or without a diagnosed concussion or between collar or noncollar groups ($P > 0.05$).

Relationships Between Metabolite Changes in the ACC and Mean g-Force of Impacts

In the ACC, the relationships between ΔtCho/tCr and head impacts with mean *g*-force thresholds of >80 *g* ($\beta = 2.7 \times 10^{-3}$), >100 *g* ($\beta = 2.6 \times 10^{-3}$), >110 *g* ($\beta = 2.5 \times 10^{-3}$), >120 *g* ($\beta = 2.3 \times 10^{-3}$), >130 *g* ($\beta = 2.1 \times 10^{-3}$), and >140 *g* ($\beta = 2.2 \times 10^{-3}$) were significantly different between participants with and without a diagnosed concussion (Fig. 4b). The relationship between ΔmyoI/tCr in the ACC and head impacts with mean *g*-force thresholds of >110 ($\beta = -1.6 \times 10^{-3}$), >120 ($\beta = -1.8 \times 10^{-3}$), and >130 *g* ($\beta = -1.3 \times 10^{-3}$) varied significantly between collar and noncollar groups. For these thresholds, the interaction effect of mean *g*-force × concussion × collar was not significant (>110 *g*: $\beta = -2.4 \times 10^{-4}$, $P = 0.18$, >120 *g*: $\beta = -2.2 \times 10^{-4}$, $P = 0.21$, >130 *g*: $\beta = -2.6 \times 10^{-4}$, $P = 0.11$), suggesting the significant effect of the collar was not confounded by concussion. Posthoc regressions revealed negative associations between ΔmyoI/tCr in the ACC and mean *g*-force of impacts >110 and >120 *g* ($\beta =$

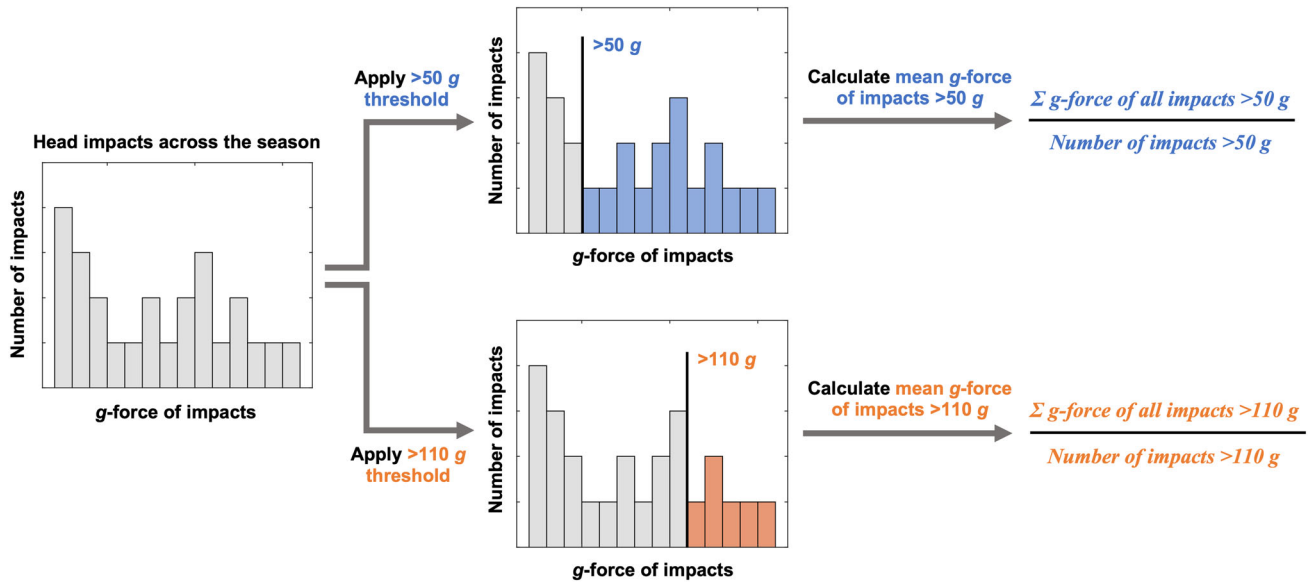

Figure 3: Calculation of mean *g*-force of head impacts using *g*-force thresholds. A schematic representation for calculation of mean *g*-force of head impacts at thresholds >50 *g* and >110 *g* are shown. Across a single season, all head impacts for each participant were recorded using an accelerometer and filtered using a hit-run filter (see Methods). Thresholds from 20 to 150 *g* in 10 *g* increments were then applied, and all impacts greater than or equal to the respective threshold are included. The mean *g*-force of head impacts was then calculated for each threshold using the sum of *g*-force of all impacts divided by the total number of impacts.

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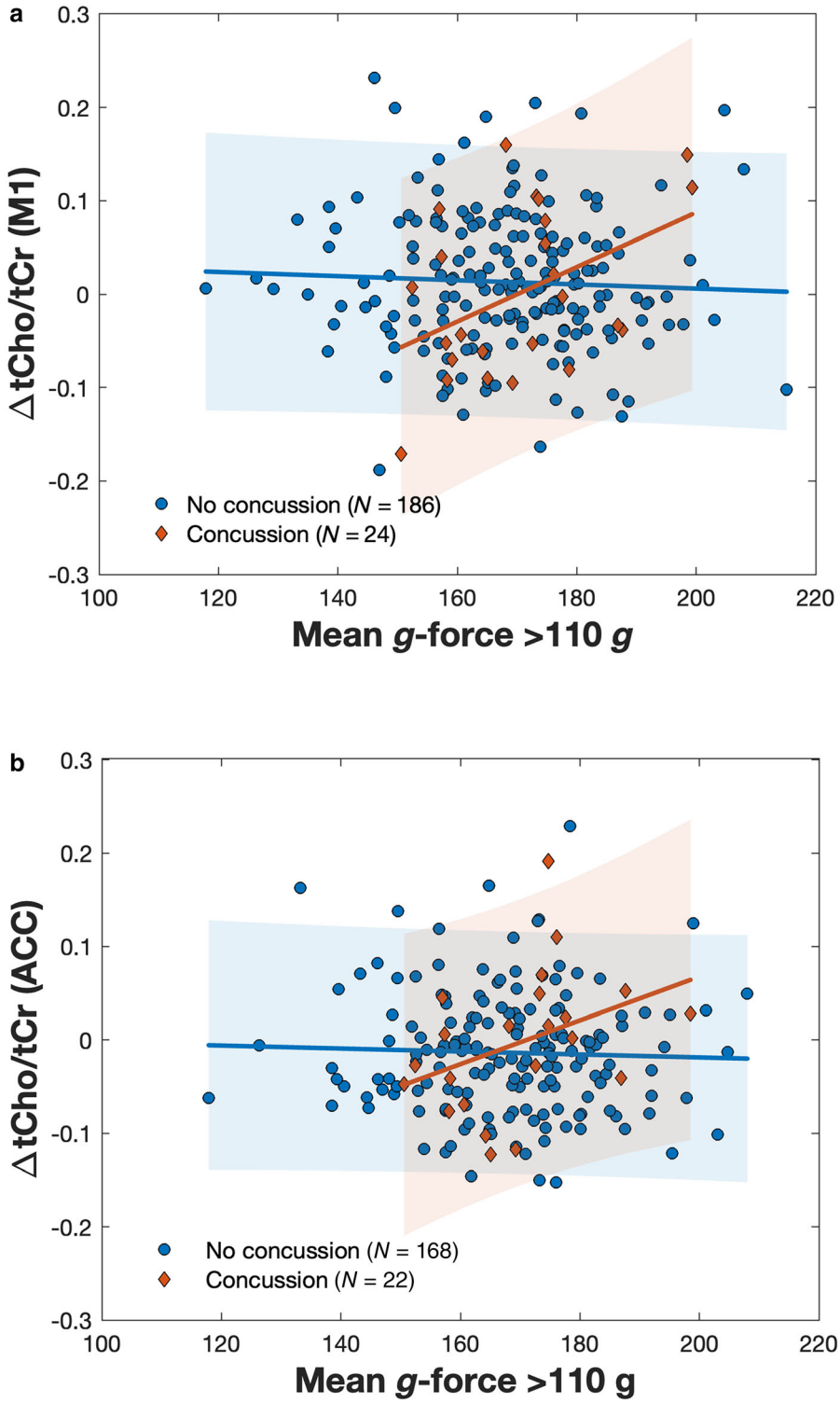

Figure 4: Univariate linear regressions between ΔtCho/tCr and mean *g*-force of impacts >110 *g*. (a) The relationship between ΔtCho/tCr and head impacts varied significantly in M1 at a mean *g*-force threshold of >110 *g* ($t = 2.68$, $P = 0.0080$) as well as for all thresholds between >80 *g* and >150 *g* (Table 2), and (b) in the ACC at a threshold of >110 *g* ($t = 2.02$, $P = 0.045$) and at thresholds of >80, >100, >120, >130, and >140 *g* (Table 2).

-1.1×10^{-3} for both thresholds) (Table 2). Concussion or the use of a collar did not significantly alter the relationship between either ΔtNAA/tCr or ΔGlx/tCr as a function of head impacts for any mean *g*-force thresholds.

Discussion

This study investigated the relationships between brain metabolites and head impacts. Across a single season of practice and games, significant differences were observed for

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Table 2. Posthoc Regressions of ΔtCho/tCr in M1 and ACC and ΔmyoI/tCr in the ACC as a Function of Mean *g*-Force of Head Impacts for Regressions With Significant Interaction Effects Between *g*-Force × Concussion or *g*-Force × Collar

Metabolites	*g*-Force Thresholds	Groups	Coefficients[a]	*t*	*P*
ΔtCho/tCr (M1)	>80 *g*	Concussion	2.3×10^{-3}	1.65	0.11
		No concussion	-4.9×10^{-4}	−1.33	0.19
	>90 *g*	Concussion	2.7×10^{-3}	1.91	6.9×10^{-2}
		No concussion	-9.6×10^{-5}	−0.27	0.79
	>100 *g*	Concussion	3.6×10^{-3}	2.85	$\mathbf{9.3 \times 10^{-3}}$
		No concussion	-1.6×10^{-4}	−0.43	0.67
	>110 *g*	Concussion	2.9×10^{-3}	2.28	$\mathbf{3.3 \times 10^{-2}}$
		No concussion	-2.2×10^{-4}	−0.66	0.51
	>120 *g*	Concussion	2.4×10^{-3}	2.00	5.8×10^{-2}
		No concussion	-2.5×10^{-4}	−0.83	0.41
	>130 *g*	Concussion	1.8×10^{-3}	1.64	0.12
		No concussion	-1.7×10^{-4}	−0.58	0.56
	>140 *g*	Concussion	1.5×10^{-3}	1.30	0.21
		No concussion	-2.9×10^{-4}	−1.01	0.31
ΔtCho/tCr (ACC)	>80 *g*	Concussion	2.3×10^{-3}	1.71	0.10
		No concussion	-3.9×10^{-4}	−1.13	0.26
	>100 *g*	Concussion	2.4×10^{-3}	1.74	0.10
		No concussion	-2.4×10^{-4}	−0.68	0.50
	>110 *g*	Concussion	2.3×10^{-3}	1.82	8.4×10^{-2}
		No concussion	-1.6×10^{-4}	−0.48	0.63
	>120 *g*	Concussion	2.1×10^{-3}	1.88	7.5×10^{-2}
		No concussion	-1.9×10^{-4}	−0.63	0.53
	>130 *g*	Concussion	1.6×10^{-3}	1.56	0.13
		No concussion	-4.2×10^{-4}	−1.44	0.15
	>140 *g*	Concussion	1.8×10^{-3}	1.82	8.4×10^{-2}
		No concussion	-3.5×10^{-4}	−1.30	0.20
ΔmyoI/tCr (ACC)	>110 *g*	Collar	-1.1×10^{-3}	−2.09	$\mathbf{3.9 \times 10^{-2}}$
		Noncollar	3.8×10^{-4}	0.65	0.51
	>120 *g*	Collar	-1.1×10^{-3}	−2.37	$\mathbf{2.0 \times 10^{-2}}$
		Noncollar	5.4×10^{-4}	1.02	0.31
	>130 *g*	Collar	-7.8×10^{-4}	−1.62	0.11
		Noncollar	3.9×10^{-4}	0.83	0.41

Significant *P*-values determined by $P \leq 0.05$ are bolded.
M1 = primary motor cortex; ACC = anterior cingulate cortex; tCr = total creatine + phosphocreatine; tCho = total phosphocholine + glycerophosphocholine; myoI = myo-inositol.
[a]Estimated coefficients, *t*-statistics, and *P*-values are calculated with univariate linear regression.

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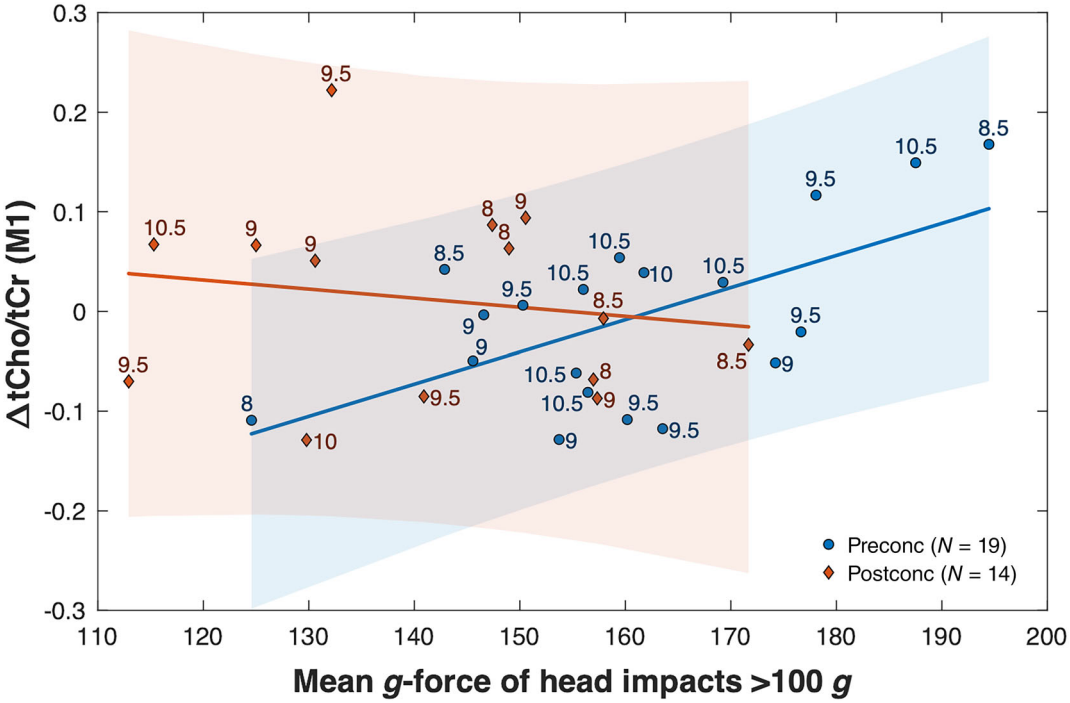

Figure 5: Relationship between ΔtCho/tCr in M1 and mean *g*-force of head impacts >100 *g* prior to and after a diagnosed concussion. Regressions are stratified by time periods spanning preseason and postconcussion ("pre-conc," blue, calculated as (postconcussion – preseason)/preseason) and postconcussion and postseason ("post-conc," orange, calculated as (postseason – postconcussion)/postconcussion). The relationship between ΔtCho/tCr and head impacts differed significantly based on the time of the concussion. Each data point is labeled with time of concussion in half month increments (i.e., 8 = first half of August, 8.5 = second half of August, etc.).

tCho/tCr and Glx/tCr between pre- and postseason for participants without a diagnosed concussion. The relationships between ΔtCho/tCr in both M1 and the ACC and head impacts varied significantly between participants with and without a diagnosed concussion at higher mean *g*-force. Post-hoc regressions revealed positive correlations between ΔtCho/tCr in M1 and mean *g*-force of impacts >100 *g* and >110 *g*, but not for any other threshold, in participants with a diagnosed concussion.

Changes in tCho/tCr Are Associated With Mean g-Force of Impacts and Vary Between Participants With and Without a Diagnosed Concussion

Increases in tCho/tCr may be associated with higher cell membrane turnover and may be regarded as an inflammatory response.[34] Associations between tCho/tCr and head impacts in the present study suggest tCho may be a key outcome metric for concussion in participants with head impacts above 80 *g*. Similarly to these observations, a prior review of 49 TBI studies reported elevated tCho levels were observed for individuals who experienced a concussion compared to healthy controls at both subacute (2 weeks–3 months) and chronic (>3 months) stages of injury.[10] In a previous cohort of 24 high school athletes, tCho/tCr was also positively associated with mean *g*-force of impacts >50 *g*, although a single time point measurement (mid-season) of tCho/tCr was used for the analysis.[22]

In participants with diagnosed concussion, the positive association between mean *g*-force of impacts >100 and >110 *g* and tCho/tCr in M1 suggest impacts with higher mean *g*-force may be accompanied by neurochemical changes in these individuals. Interestingly, the relationship between changes in tCho/tCr in M1 and head impacts recorded after a diagnosed concussion differed from impacts recorded before concussion for mean *g*-force of impacts >100 *g*, suggesting the effect of concussion on neurochemical alterations may also be observed on a timeframe from weeks to months. Together, these results suggest a concussion after repeated impacts may produce a distinct neurochemical response compared to repeated impacts in individuals without a concussion during a single season.

tNAA/tCr Did Not Vary as a Function of Time, Concussion, Impacts, or the Use of a Collar

A prior study of neurochemical changes in response to head impacts or concussion have focused on changes in NAA as a metric for neuronal and axonal health[35]; however, this study did not observe significant differences in tNAA/tCr as a function of time, concussion, or the use of a collar. Decreases in tNAA in acute and chronic phases postinjury have been consistently observed in collegiate and professional football athletes, suggesting metabolite markers of injury may be age

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dependent.[9,36] This may be due to continuous maturation of white matter throughout adolescence into adulthood,[37] as the concentration of tNAA in white matter increases across the course of adolescence.[38] It is possible the changes in tNAA in adolescents are too small to detect with MRS due to lower tNAA concentrations compared to adults. Similarly, tNAA in the adolescent brain may be less sensitive to injury. This study did not observe a significant correlation between ΔtNAA/tCr and head impacts at any *g*-force threshold. This is consistent with prior work exploring subconcussive head impacts in high school football athletes, where tNAA did not vary across the season and no correlations were observed between tNAA and head impacts.[22]

MyoI/tCr Exhibit Longitudinal Reduction in Participants Without a Diagnosed Concussion

Prior studies of contact sports athletes without a concussion reported longitudinal decreases in myoI.[20,21] For example, one study observed lower myoI in the dorsolateral prefrontal cortex (DLPFC) during mid- and postseason compared to baseline in a cohort of high school American football athletes.[21] Previous work in a cohort of collegiate basketball players without a diagnosed concussion also reported lower myoI/tCr in the frontal white matter postseason compared to both pre- and mid-season.[20] In the present study, a pre- to postseason reduction in myoI/tCr was observed in the ACC but did not survive FDR correction. Conclusions of myoI changes are mixed, and while some studies suggest it may be a marker of subconcussive injury, further investigation is required.

Glx/tCr Vary Between Pre- and Postseason in Participants Without a Diagnosed Concussion

Reported longitudinal changes in Glx after TBI are, overall, inconsistent. Prior work reported a longitudinal decrease in Glx in M1 in high school football athletes[21]; however, this study did not observe significant changes in Glx in M1 as a function of time, concussion, or use of a JVC collar. Another study of high school American football athletes with no diagnosed concussion across a single season reported a significant reduction in Glx in the DLPFC at mid-season relative to preseason, and a significant increase at postseason compared to mid-season.[22] In the present study, no pre- to postseason changes in Glx were observed, suggesting Glx may reflect metabolite changes in a time frame shorter than a full season, and acquiring MRS scans one or 2 months apart may be required to capture dynamics of Glx fluctuations after subconcussive injury. These results also suggest differences in voxel location, particularly white and grey matter fractions, and imaging schemes may contribute to reported differences in Glx changes after brain injury.

The Effect of a JVC Collar

Prior preclinical and clinical investigations studying the use of a JVC collar on diffusion tensor imaging (DTI) metrics and histopathological evaluations after concussion reported significant changes in pre- and postinjury metrics in noncollar groups, whereas these changes were not observed in collar groups, suggesting the use of a JVC collar may provide protection to the brain after injury.[11,24,39] In this study, the relationship between ΔmyoI/tCr in the ACC and mean *g*-force of impacts >110, >120, and >130 *g* significantly varied with use of a collar. Posthoc analysis in the collar group revealed ΔmyoI/tCr in the ACC was negatively associated with mean *g*-force of impacts >110 and >120 *g*. Interestingly, at lower mean *g*-force of impacts, myoI was generally higher at postseason compared to preseason. At an inflection point of ∼170 g, the trend shifts to a reduction in myoI postseason relative to preseason. MyoI is an osmolyte found in membranes of glial cells, which are involved in the biochemical cascade of neuroinflammation after brain injury, and may promote persistent release of proinflammatory factors that can lead to further disturbance in cellular metabolism to different regions of the brain.[10,13,40] The relative longitudinal decrease in myoI/tCr associated with repeated impacts at higher magnitudes (eg, >170 *g*) may be indicative of a reduction of neuroinflammation in the presence of the collar, which may not be present at lower impact magnitudes. Prior work using the same cohort of athletes without a diagnosed concussion also reported a significant interaction effect for collar × head impacts for the relationships between pre- to postseason changes in DTI metrics and head impacts.[24] Opposite trends between collar and noncollar groups were observed for these DTI metrics as a function of impacts at higher magnitudes (all thresholds between >110 and >140 *g*).[24] The effect of the JVC collar may be tightly linked to head impact magnitude and more prominent at higher *g*-force. The difference in association between ΔmyoI/tCr and mean *g*-force of impacts between collar and noncollar groups, combined with prior observations of longitudinal changes in DTI metrics only observed in athletes in the noncollar group, suggest the use of a JVC collar may protect against a subset of neurological alterations after head injury.

Limitations

All data in this study were acquired using scanners from the same vender and at the same field strength. Only male players were included in this study of football players. The effect of age was not considered, but all athletes were adolescents. The studied population only included American football athletes with head impacts. Comparison with an age-matched adolescent control group without head impacts may be considered in future work to control for fluctuations in metabolites over time (eg, summer to fall). We acknowledge some of our key results focus on differences between those with and without a

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diagnosed concussion. Given the subjective nature of concussion diagnosis, one explanation is our results may explain differences in symptomatic head impacts compared to those without symptoms (who may not be diagnosed with a concussion due to lack of symptoms). While mean *g*-force is harder to interpret than number of impacts, mean *g*-force accounts for both magnitude and number of head impacts. There were also no clear trends in the median number of impacts >80 *g*, supporting the use of mean *g*-force as a metric for head impacts as higher mean *g*-force values were not dominated by isolated incidents or a few high impact hits. Importantly, no significant relationships were observed for impacts with mean *g*-force thresholds less than 80 *g* for any metabolites, as lower thresholds included impacts encompassing a wider range of magnitudes as well as reduced sensitivity of the accelerometer to lower forces. Future studies would benefit from evaluation of both linear and rotational impact forces. Lastly, MRS data were acquired largely in cortical regions, which were selected due to the mechanism of action of the JVC collar to impact cerebral blood flow and time available within the MR examination. Consideration of metabolite changes in predominantly white matter regions, as well as absolute metabolite concentrations (rather than relative ratios) is an important area of future investigation.

Conclusion

This study observed the relationships between ΔtCho/tCr and head impacts at higher mean *g*-force thresholds varied significantly between participants with and without a diagnosed concussion, suggesting tCho/tCr may be a key outcome metric for concussion as well as head impacts. The relationship between ΔmyoI/tCr and mean *g*-force of impacts also varied between collar and noncollar groups, and ΔmyoI/tCr was significantly and negatively correlated with impacts with mean *g*-force thresholds of >110 and >120 *g*. The present study includes the largest longitudinal cohort of American football athletes to date and may be generalizable to adolescent athletes exposed to repeated head impacts and concussions. Our results build upon prior sports injury research demonstrating repeated head impacts and concussions induce distinctive neurochemical responses and support the continued use of neuroimaging metrics in characterizing these changes.

Author Contributions

J.A. Dudley, J.A. Diekfuss, W.Y., K.M.C., G.D.M., C.C.F. engaged in research conceptualization; J.A. Dudley, J.A. Diekfuss, W.Y., K.D.B.F., G.D.M., C.C.F. developed methodology; J.A. Dudley, J.A. Diekfuss, K.D.B.F., G.D.M. collected data; Z.L., J.A. Dudley, J.A. Diekfuss, N.A., K.D.B.F., G.D.M. managed data for interpretation; Z.L., J.A. Dudley, W.Y. developed and implemented software; Z.L., J.A. Dudley, N.A., C.C.F conducted formal data analysis; Z.L., J.A. Dudley, W.Y., C.C.F. prepared data visualization; J.A. Diekfuss, G.D.M., C.C.F. supervised and coordinated research activity planning and execution; Z.L., C.C.F. prepared initial draft; J.A. Dudley, J.A. Diekfuss, N.A., A.D.E., K.M.C., W.Y., T.M.Z., A.B.S.-G., K.D.B.F., G.D.M., C.C.F. reviewed and edited final drafts.

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